

YUXIANG JIANG

New York, NY | 646-306-6292 | yj548@cornell.edu | yuxiangjiangct.github.io/portfolio

EDUCATION

Cornell Tech, M.S. Computer & Information Sciences | *New York, NY*

Aug 2024 – Expected May 2026

Jinan University, B.E. Internet of Things Engineering | *Guangzhou, China*

Sep 2020 – Jul 2024

TECHNICAL SKILLS

Languages: Java, Python, C++, TypeScript/JavaScript, HTML/CSS, SQL, Go

Backend: Spring Boot, FastAPI, Node.js/Express, REST/gRPC, Microservices

Systems: Data Structures, Distributed Systems, Concurrency, Linux/Unix, Networking (TCP/IP, HTTP)

Cloud/DevOps & Testing: AWS, Docker, CI/CD, Observability; JUnit, PyTest, API & Load Testing

Data/Storage: PostgreSQL, MySQL, Redis, MongoDB; Schema Design, Indexing & Query Optimization

EXPERIENCE

Dataman Analytics (quantdefi.ai), Software Engineering Intern | *New York, NY*

Jun 2025 – Aug 2025

- Owned and re-architected a FastAPI microservices platform, cutting p99 latency by ~70% and doubling peak RPS with cache-aside Redis, connection pooling, and async I/O.
- Led standardization of the pool scanner using a third-party data provider (DeFiLlama) and shipped it as a FastAPI service with a Streamlit dashboard and opportunity alerts so non-engineers can pull insights without engineering support.
- Designed and shipped an ARIMA-based APY forecasting pipeline for 19,000+ pools and a market-regime-adaptive ranking powering bots/newsletters.
- Established end-to-end observability (structured logs, request IDs, dashboards, p95/p99, SLO-based Slack alerts) and hardened reliability with JWT, ECDSA signatures, rate limiting, and idempotency keys.

PolyPoll (AI-driven Event Intelligence Platform), Founding Engineer | *New York, NY*

Jul 2025 – Present

- Architected a multi-provider inference routing layer (Python/FastAPI) with automatic failover across 6 model endpoints via Groq/OpenRouter.
- Enforced rate limits and daily quota budgets, validating reliability via k6 soak tests (~500 VUs, 28K+ requests, <0.5% errors).
- Built a market generation engine via a 10-stage pipeline with dual-layer semantic de-duplication (0.75 similarity threshold + 48h window) to filter near-duplicate markets and improve quality.
- Integrated 5+ external APIs (Exa, Perplexity, Replicate, Cloudinary, Slack) with entity resolution against a database of 1,700+ public figures, enabling graceful degradation, retry logic, and unified cost monitoring.
- Designed PostgreSQL schema with auto-categorization triggers, RLS policies, and JSONB embeddings for efficient market discovery and similarity matching.

PROJECTS

bounties.fun (ETHGlobal – Worldcoin Prize Winner) | *FastAPI, PostgreSQL/Supabase, Web3.py, World ID MiniKit*

- Integrated World ID zero-knowledge proof verification (MiniKit), preventing Sybil attacks via nullifier_hash uniqueness.
- Built a hybrid verification architecture combining real-time Twitter stream monitoring (<30s payout) with a daily batch pipeline using cursor pagination to handle API rate limits.
- Implemented a concurrency-safe payout system with optimistic locking in PostgreSQL, ensuring atomic payout execution.

Scalable URL Shortener & Analytics | *Node.js, Redis, PostgreSQL, Docker*

- Built a URL shortener handling 500+ RPS with p95 <50ms via Redis cache-aside and NanoID.
- Batched click events with a Redis queue, cutting PostgreSQL write I/O by 90%.
- Containerized the stack with Docker Compose and added rate limiting and input sanitization.

Postgraduate Recommendation & Evaluation System | *Spring Boot, React, MyBatis, MySQL, Redis, AWS*

- Automated the postgraduate recommendation workflow for 500+ faculty and students, eliminating 90% of paper and email steps.
- Optimized MyBatis queries and introduced Redis caching, cutting average page load by 70% (400ms → 120ms) and boosting form-submission success to 99%.
- Orchestrated CI/CD and zero-downtime releases on AWS EC2 using CodePipeline for reliable production deployment.